

CHAPTER-WISE PROBLEMS

Probability Distributions

Statistical Foundation of Data Science
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1 Bernoulli Distribution

Problem 1: Basic Bernoulli Properties and Moments

A biased coin has probability $p = 0.65$ of showing heads (success). Let X be the Bernoulli random variable indicating the outcome.

- (a) Write the probability mass function (PMF) of X :

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Compute $P(X = 0)$ and $P(X = 1)$ for $p = 0.65$.

- (b) Derive the expected value $\mathbb{E}[X]$ using the definition:

$$\mathbb{E}[X] = \sum_x x \cdot P(X = x)$$

- (c) Derive the variance $\text{Var}(X)$ using both methods:

- Method 1: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
- Method 2: Direct formula $\text{Var}(X) = p(1 - p)$

Verify both methods give the same result for $p = 0.65$.

- (d) Calculate the standard deviation $\sigma = \sqrt{\text{Var}(X)}$. At what value of p is the variance maximized? Prove this by taking $\frac{d}{dp} \text{Var}(X) = 0$.

Problem 2: Bernoulli Process and Geometric Distribution

A quality inspector tests manufactured items where each item has probability $p = 0.15$ of being defective (independent trials).

- (a) If the inspector tests 5 items independently, write the probability that exactly 2 are defective using Bernoulli trials:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_5 = x_5) = \prod_{i=1}^5 p^{x_i} (1 - p)^{1-x_i}$$

Calculate for the sequence: Defective, Good, Defective, Good, Good.

- (b) Let Y be the number of items tested until the first defective is found (geometric distribution). The PMF is:

$$P(Y = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

Compute $P(Y = 1)$, $P(Y = 2)$, and $P(Y \leq 3)$.

- (c) Derive the expected number of trials until first success:

$$\mathbb{E}[Y] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

Use the formula for the sum of a geometric series: $\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$.

Show that $\mathbb{E}[Y] = \frac{1}{p}$ and compute for $p = 0.15$.

- (d) Calculate the probability that the inspector needs to test more than 10 items to find the first defective:

$$P(Y > 10) = (1-p)^{10}$$

Problem 3: Maximum Likelihood Estimation for Bernoulli

A data scientist observes $n = 8$ independent Bernoulli trials with outcomes: $X_1 = 1, X_2 = 0, X_3 = 1, X_4 = 1, X_5 = 0, X_6 = 1, X_7 = 1, X_8 = 1$.

- (a) Write the likelihood function:

$$L(p) = \prod_{i=1}^8 p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i}$$

Calculate $\sum x_i$ and express $L(p)$ explicitly.

- (b) Write the log-likelihood function:

$$\ell(p) = \log L(p) = \left(\sum x_i\right) \log p + \left(n - \sum x_i\right) \log(1-p)$$

- (c) Find the maximum likelihood estimator (MLE) by solving:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0$$

Show that $\hat{p}_{\text{MLE}} = \frac{\sum x_i}{n}$ (sample proportion).

- (d) Verify this is a maximum by checking the second derivative:

$$\frac{d^2\ell}{dp^2} = -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1-p)^2}$$

Show that this is negative at $\hat{p}_{\text{MLE}}$, confirming a maximum. Compute the MLE numerically for the given data.

2 Binomial Distribution

Problem 4: Binomial Probability Calculations

A multiple-choice exam has 10 questions, each with 4 options. A student guesses randomly on all questions. Let X be the number of correct answers, so $X \sim \text{Binomial}(n = 10, p = 0.25)$.

(a) Write the binomial PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

Compute $P(X = 3)$ exactly.

(b) Calculate the probability of passing (at least 6 correct):

$$P(X \geq 6) = \sum_{k=6}^{10} \binom{10}{k} (0.25)^k (0.75)^{10-k}$$

(c) Compute the expected number of correct answers:

$$\mathbb{E}[X] = np$$

and the variance:

$$\text{Var}(X) = np(1 - p)$$

Calculate the standard deviation $\sigma = \sqrt{\text{Var}(X)}$.

(d) Use the normal approximation to estimate $P(X \geq 6)$. Apply continuity correction:

$$P(X \geq 6) \approx P\left(Z \geq \frac{5.5 - \mu}{\sigma}\right)$$

where $Z \sim \mathcal{N}(0, 1)$, $\mu = \mathbb{E}[X]$, and $\sigma = \sqrt{\text{Var}(X)}$. Compare with the exact binomial calculation from part (b).

Problem 5: Binomial Distribution as Sum of Bernoulli Trials

Let $X_1, X_2, \dots, X_n$ be independent Bernoulli(p) random variables, and $S_n = \sum_{i=1}^n X_i$.

(a) Show that $S_n \sim \text{Binomial}(n, p)$ by computing the probability:

$$P(S_n = k) = P\left(\sum_{i=1}^n X_i = k\right)$$

Use the fact that there are $\binom{n}{k}$ ways to choose which k trials result in success, and each specific sequence has probability $p^k(1 - p)^{n-k}$.

(b) Using the linearity of expectation, show that:

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = np$$

(c) For independent random variables, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. Use this to show:

$$\text{Var}(S_n) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p)$$

(d) For $n = 20$ and $p = 0.4$, compute:

- The mode (most likely value) of the distribution
- The probability that S_n is within one standard deviation of its mean: $P(\mu - \sigma \leq S_n \leq \mu + \sigma)$

Problem 6: Hypothesis Testing with Binomial

A pharmaceutical company claims their drug has a success rate of $p = 0.70$. In a clinical trial with $n = 25$ patients, only $k = 14$ show improvement.

(a) State the null and alternative hypotheses:

$$H_0 : p = 0.70 \quad \text{vs} \quad H_1 : p < 0.70$$

Under H_0 , $X \sim \text{Binomial}(25, 0.70)$.

(b) Calculate the p-value: the probability of observing $k = 14$ or fewer successes under H_0 :

$$\text{p-value} = P(X \leq 14 | p = 0.70) = \sum_{k=0}^{14} \binom{25}{k} (0.70)^k (0.30)^{25-k}$$

(c) Using the normal approximation with continuity correction:

$$\text{p-value} \approx P\left(Z \leq \frac{14.5 - np}{\sqrt{np(1-p)}}\right)$$

where $Z \sim \mathcal{N}(0, 1)$. Compute the z-score and find the p-value from the standard normal table.

(d) At significance level $\alpha = 0.05$, should we reject H_0 ? Construct a 95% confidence interval for p using the normal approximation:

$$\hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $\hat{p} = 14/25$. Does this interval contain 0.70?

3 Poisson Distribution

Problem 7: Poisson Process and Probability Calculations

Customers arrive at a bank at an average rate of $\lambda = 4$ per hour. Let X be the number of arrivals in one hour, where $X \sim \text{Poisson}(\lambda = 4)$.

- (a) Write the Poisson PMF:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Compute $P(X = 0)$, $P(X = 2)$, and $P(X = 5)$.

- (b) Calculate the probability that at most 3 customers arrive:

$$P(X \leq 3) = \sum_{k=0}^3 \frac{e^{-4} \cdot 4^k}{k!}$$

- (c) The mean and variance of a Poisson distribution are both equal to λ . Verify that:

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda$$

Calculate the standard deviation for $\lambda = 4$.

- (d) If the observation period changes to 30 minutes (0.5 hours), what is the new parameter? Let Y be the number of arrivals in 30 minutes. Show that $Y \sim \text{Poisson}(\lambda' = 2)$ and compute $P(Y = 1)$.

Problem 8: Poisson Approximation to Binomial

When n is large and p is small such that $np = \lambda$ is moderate, $\text{Binomial}(n, p) \approx \text{Poisson}(\lambda)$.

A factory produces 1000 items per day, and each has a 0.003 probability of being defective. Let X be the number of defective items.

- (a) Exactly, $X \sim \text{Binomial}(n = 1000, p = 0.003)$. Write the binomial formula for $P(X = k)$ (don't compute yet).
- (b) Calculate $\lambda = np = 1000 \times 0.003 = 3$. Use the Poisson approximation $X \approx \text{Poisson}(\lambda = 3)$ to compute:

$$P(X = 2) \approx \frac{e^{-3} \cdot 3^2}{2!}$$

- (c) Compare the Poisson approximation with the exact binomial calculation:

$$P(X = 2) = \binom{1000}{2} (0.003)^2 (0.997)^{998}$$

(Use the approximation $(0.997)^{998} \approx e^{-3}$ and $\binom{1000}{2} = 499500$ to simplify.)

- (d) Compute $P(X \geq 5)$ using the Poisson approximation:

$$P(X \geq 5) = 1 - P(X \leq 4) = 1 - \sum_{k=0}^4 \frac{e^{-3} \cdot 3^k}{k!}$$

Problem 9: Sum of Independent Poisson Variables

The reproductive property of Poisson: If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent, then $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

A call center has two phone lines. Line 1 receives calls at rate $\lambda_1 = 3$ per hour, and Line 2 at rate $\lambda_2 = 5$ per hour.

- (a) Let X and Y be the number of calls on Lines 1 and 2 in one hour, respectively. Write the distributions: $X \sim \text{Poisson}(3)$, $Y \sim \text{Poisson}(5)$.
- (b) Let $Z = X + Y$ be the total number of calls. State the distribution of Z and compute $P(Z = 6)$ using:

$$Z \sim \text{Poisson}(\lambda_1 + \lambda_2 = 8)$$

$$P(Z = 6) = \frac{e^{-8} \cdot 8^6}{6!}$$

- (c) Verify the reproductive property by computing $P(Z = 6)$ directly from the convolution formula:

$$P(Z = 6) = \sum_{k=0}^6 P(X = k)P(Y = 6 - k) = \sum_{k=0}^6 \frac{e^{-3} \cdot 3^k}{k!} \cdot \frac{e^{-5} \cdot 5^{6-k}}{(6-k)!}$$

Show that this simplifies to the Poisson(8) formula.

- (d) Calculate the expected total number of calls $\mathbb{E}[Z]$ and variance $\text{Var}(Z)$ using:

$$\mathbb{E}[Z] = \mathbb{E}[X] + \mathbb{E}[Y], \quad \text{Var}(Z) = \text{Var}(X) + \text{Var}(Y)$$

(assuming independence).

4 Gaussian (Normal) Distribution

Problem 10: Standard Normal Calculations and Transformations

Let $Z \sim \mathcal{N}(0, 1)$ be a standard normal random variable.

- (a) Write the probability density function (PDF):

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Calculate $f_Z(0)$, $f_Z(1)$, and $f_Z(-1)$. Verify the symmetry property: $f_Z(-z) = f_Z(z)$.

- (b) Using the standard normal table ($\Phi(z) = P(Z \leq z)$), compute:

- $P(Z \leq 1.96)$
- $P(Z > 1.5)$

- $P(-1 \leq Z \leq 2)$
- $P(|Z| > 2)$ (two-tailed probability)

(c) Find the critical values (z-scores):

- $z_{\alpha/2}$ such that $P(Z > z_{\alpha/2}) = 0.025$ (for 95% confidence)
- z_{α} such that $P(Z > z_{\alpha}) = 0.05$ (for 90% confidence, one-tailed)

(d) Let $X \sim \mathcal{N}(\mu = 50, \sigma^2 = 16)$. Transform X to standard normal using:

$$Z = \frac{X - \mu}{\sigma}$$

Calculate $P(X > 58)$ by converting to a Z -score and using the standard normal table.

Problem 11: Normal Distribution Properties and Calculations

IQ scores are normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$. Let X denote an individual's IQ score.

(a) Write the PDF of $X \sim \mathcal{N}(100, 15^2)$:

$$f_X(x) = \frac{1}{15\sqrt{2\pi}} \exp\left(-\frac{(x - 100)^2}{2(15)^2}\right)$$

(b) Calculate the probability that a randomly selected person has an IQ between 85 and 115 (within one standard deviation of the mean):

$$P(85 \leq X \leq 115) = P\left(\frac{85 - 100}{15} \leq Z \leq \frac{115 - 100}{15}\right) = P(-1 \leq Z \leq 1)$$

(c) Find the 90th percentile of IQ scores: the value $x_{0.90}$ such that $P(X \leq x_{0.90}) = 0.90$. Use:

$$x_{0.90} = \mu + z_{0.90}\sigma$$

where $z_{0.90} = 1.282$ (from standard normal table).

(d) A "genius" is defined as having IQ above 130. What proportion of the population qualifies? Calculate:

$$P(X > 130) = P\left(Z > \frac{130 - 100}{15}\right)$$

If 10,000 people are tested, how many are expected to have IQ above 130?

Problem 12: Linear Combinations and Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be independent random variables with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$.

- (a) Show that any linear combination is also normal:

$$Y = a_1X_1 + a_2X_2 + \dots + a_nX_n \sim \mathcal{N}\left(\sum_{i=1}^n a_i\mu_i, \sum_{i=1}^n a_i^2\sigma_i^2\right)$$

Verify this for $n = 2$: If $X \sim \mathcal{N}(3, 4)$ and $Y \sim \mathcal{N}(5, 9)$ are independent, find the distribution of $Z = 2X + 3Y$.

- (b) The sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ where $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. Show that:

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

For $\mu = 50$, $\sigma = 10$, and $n = 25$, compute $P(\bar{X} > 52)$.

- (c) State the Central Limit Theorem: If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 (from any distribution), then for large n :

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Explain why this is crucial for statistical inference.

- (d) An exponential random variable has mean $\mu = 5$ and variance $\sigma^2 = 25$. If we take a sample of $n = 100$ observations and compute the sample mean $\bar{X}$, approximate:

$$P(4.5 \leq \bar{X} \leq 5.5)$$

using the CLT (even though the original distribution is not normal).